Tidal potential measured on a coastal–inland optical-clock baseline

Abstract — draft 1

Tides deform the solid Earth and redistribute ocean water, changing the gravitational potential and the relative ticking rates of distant clocks. Optical-clock comparisons can therefore test the ability of precision timekeeping to measure a changing gravitational environment. Here we detect tidal variations in the potential difference between coastal Shanghai and inland Wuhan using strontium and ytterbium clocks 692 km apart. From 545.9 hours of observations spanning 353 days, we obtain an overall tidal response of 0.96 ± 0.11 relative to the combined solid-Earth and ocean-loading prediction. We further distinguish neighbouring twice-daily tidal responses using the changing lunar–solar phase across the interrupted record. With relative amplitudes and phases fixed within each group, the lunar M_2+N_2 response is 1.00 ± 0.13 and the neighbouring S_2+K_2 response is 0.93 ± 0.36 (one statistical standard error). The lunar response agrees with the predicted phase; the neighbouring group, which includes the principal solar tide, is more sensitive to disturbances at 12 hours. These observations extend optical-clock measurements to dynamic potential differences and demonstrate the separation of neighbouring tidal responses on an intermittently operated baseline.